

Hackathon Problem Statement

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Title

Smart Doctor Appointment & Clinic Management Platform

Technology Stack

Frontend

- React.js (Mandatory)

Backend

Any one of:

- Golang (Echo Framework)
- Python (FastAPI / Django)
- Node.js (Express.js / NestJS)

Database

Any one of:

- PostgreSQL
 - MySQL
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Problem Overview

Patients often face difficulties in finding doctors, booking appointments, managing schedules, and tracking appointment status.

Build a web-based platform that allows patients to search doctors, book appointments, manage schedules, and enables administrators to manage doctors, patients, and appointments.

The platform should support both:

1. Admin Portal

Functional Requirements

Admin Module

Doctor Management

Admin should be able to:

- Create Doctor
- Update Doctor
- Delete Doctor
- Activate/Deactivate Doctor

Doctor Fields:

- Name
 - Specialization
 - Experience
 - Qualification
 - Consultation Fee
 - Available Days
 - Available Time Slots
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Appointment Management

Admin should be able to:

- View all appointments
- Approve appointments
- Cancel appointments
- Mark appointment as completed

Statuses:

- Pending
- Confirmed
- Cancelled

- Completed
-

Dashboard

Display:

- Total Doctors
 - Total Patients
 - Today's Appointments
 - Upcoming Appointments
 - Completed Appointments
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Patient Module

Registration & Login

Patient should be able to:

- Register
 - Login
 - Logout
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Doctor Search

Search by:

- Doctor Name
- Specialization

Filter by:

- Fee Range
 - Available Date
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Appointment Booking

Patient should be able to:

- Select Doctor
 - Choose Date
 - Choose Time Slot
 - Book Appointment
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My Appointments

Patient should be able to:

- View appointments
 - Cancel appointment
 - View appointment status
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Edge Cases

Double Booking Prevention

Same slot should not be booked twice.

Example:

Doctor A
10:00 AM - 10:30 AM

If booked by Patient X,
Patient Y must not see the slot.

Past Date Booking

User should not be allowed to:

- Book yesterday
 - Book previous time slots
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Doctor Deactivation

If a doctor is inactive:

- New appointments cannot be booked
 - Existing appointments remain visible
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Cancellation Rules

Cannot cancel:

- Completed appointment

Can cancel:

- Pending
 - Confirmed
-

Duplicate Registration

Prevent duplicate accounts using:

- Email
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Concurrent Requests

If two users attempt booking the same slot:

System must ensure only one booking succeeds.

Bonus Features

- JWT Authentication
- Email Notifications

- Doctor Profile Photo Upload
 - Appointment Calendar View
 - Pagination
 - Search Optimization
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Evaluation Criteria

Criteria	Weightage
Architecture	20%
API Design	20%
Frontend UX/UI	20%
Database Design	15%
Edge Case Handling	15%
Code Quality	10%

Deliverables

1. Doctors can create appointments.
2. Patients should book an appointment through this portal.

Participants must submit:

- Source Code (GitHub REPO)
- Database Schema
- Screenshots

The application should be runnable locally with documented setup instructions.